COURSE FINAL PROJECT · IMPLEMENTATION REPORT

Airfoil Explorer

DRL × LBM Design Space

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SECOND PARTICIPANT
Not supplied

DATE
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STATUS
Live product deployed and verified

Problem and required achievement

The thesis workflow uses lattice Boltzmann analysis to evaluate airfoils and deep reinforcement learning to explore BP3333 geometry changes. The result set is scientifically useful but difficult to review interactively: geometry and aerodynamic coefficients must stay synchronized, while fields not approved for public release remain outside the deployed artifact.

Defined achievement

A public reviewed-data application in which a reviewer drags geometry parameters, sees the nearest stored wing and its C_l, C_d, and C_l/C_d values update live, and releases to snap every control to that exact row. The public contract is limited to declared geometry, coefficients, physical conditions, and minimal provenance. The interface never predicts or interpolates aerodynamic values.

Why this is an engineering task

- **Scientific consistency:** the wing and every displayed coefficient must originate from one admitted provenance row.
- **Interaction quality:** nearest-neighbor search must remain deterministic, fast, accessible, and stable during drag.
- **Publication safety:** credentials, local paths, W&B namespace, and undeclared diagnostics remain outside the public artifact.
- **Operational safety:** publication must fail closed if export, validation, privacy scanning, or deployment fails.

Safety model

The selected delivery target is a **public GitHub Pages** project site. An approved workflow runs on every `main` push, hourly, or manually: it retrieves only the declared W&B history keys, validates the complete snapshot, scans the exact public artifact, and deploys only after every gate passes. The synthetic fixture remains available for local QA but has no public deployment path. URL: edengol10.github.io/AI-final-course/. References: GitHub Pages workflows and W&B Public API.

Repository boundary. The thesis checkout remained a read-only scientific reference at commit `4936dea`; production code neither imports nor mutates it. All implementation lives in the separate `AI-final-course` repository.

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Product and architecture

A static, credential-free client backed by validated, immutable data snapshots.

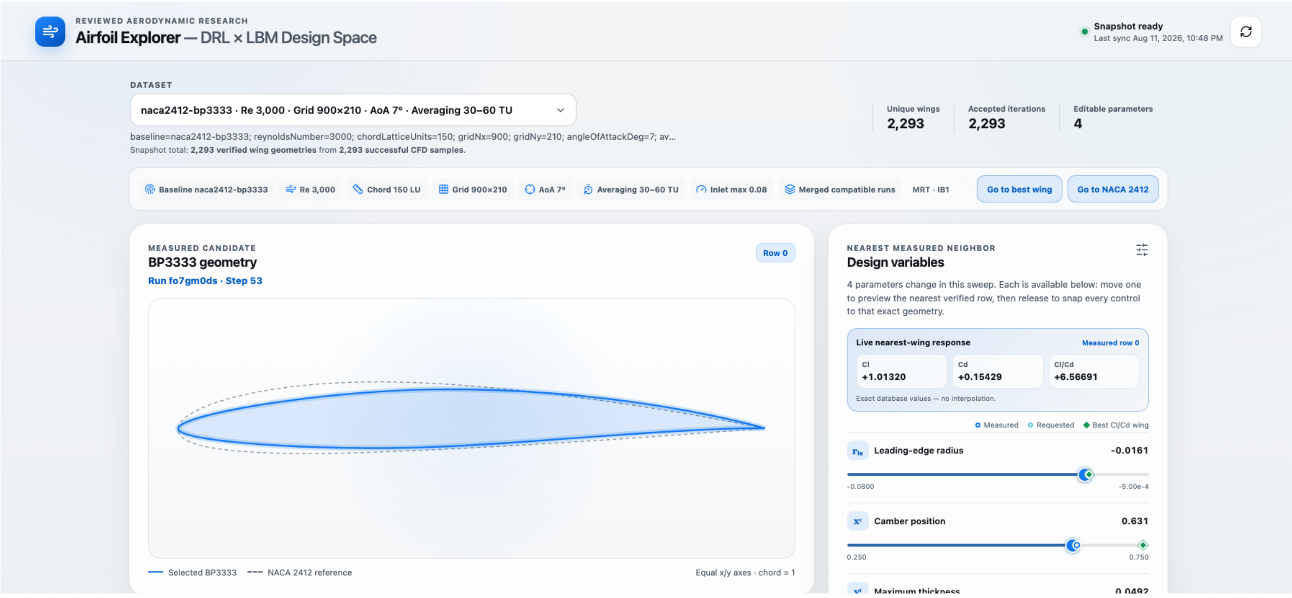

Figure 1. Deployed Airfoil Explorer loading the reviewed W&B snapshot.

2,293 verified wing geometries	**2,293** successful CFD samples	**5** reviewed run identities	**1** compatible condition group

Data path and trust boundary

Approved W&B runs → narrow per-iteration history export → physical grouping and admission → canonical JSON/SHA-256 → Vite build and privacy scan → GitHub Pages → browser Zod/hash validation and Web Worker search.

Interaction contract

- During pointer or keyboard drag, an animation-frame-throttled worker normalizes only variable parameters by authoritative BP3333 bounds and selects the exact nearest row.
- Ghost markers preserve requested positions; the wing, signed C_l/C_d bars, and provenance preview always change together.
- On release, all sliders animate to the selected database vector. Stable record index breaks equal-distance ties, preventing flicker.
- Refresh recomputes the manifest's canonical SHA-256, then each shard's byte size and SHA-256, and validates every schema before an atomic in-memory swap. Unchanged, malformed, stale, offline, and empty states remain distinct.

Public data contract

The deployed manifest is explicitly `reviewed-wandb`. The approved scientific outputs are BP3333 geometry/parameters, C_l, C_d, and the computed C_l/C_d ratio. Minimal run/step provenance, physical-condition descriptors, bounds, counts, and checksums remain for auditability. Credentials, W&B entity/project names, private machine paths, undeclared diagnostics, and per-row coordinate arrays are absent.

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AI co-coding process and skill evidence

AI accelerated implementation, while domain contracts and human review controlled scientific risk.

Same-prompt before/after experiment

> **"Build the W&B-to-dashboard data refresh and nearest-wing interaction while preserving scientific validity and protecting unpublished data."**

A prospective unskilled design was frozen before the custom skill was read. The identical prompt was then answered with the *curate-aero-wandb-data* skill. Each output was scored 0–5 against six predeclared criteria; both artifacts and the rubric are preserved in `experiments/skill-comparison/`.

Criterion	Without	With	Observed correction
Secret exposure	4/5	5/5	Exact narrow-history and exclusion boundary
Per-wing keys	2/5	5/5	Pinned step C_l/C_d values
Invalid rejection	3/5	5/5	Ordered, stable reason codes
Physical grouping	3/5	5/5	Exact dimensions; unknowns isolated by run
Provenance	4/5	5/5	Minimal run/step/time tied to one metric tuple
Reproducibility	3/5	5/5	Authoritative bounds and deterministic representative
Total	**19/30**	**30/30**	**Domain rules became testable contracts**

Material human corrections

- Rejected observed-range normalization: the nearest wing would change when sample coverage changed. Fixed authoritative BP3333 bounds now define distance.
- Reduced the published schema to the explicitly approved geometry and coefficient fields.
- Removed an early per-row coordinate-array field; TypeScript reconstructs 253 geometry points from the ten parameters.
- Replaced an analytic reference curve with the exact BP3333 NACA2412 baseline; split unique-wing and successful-sample counts.

Skills and selected professional prompts

The final project methodology uses only the following three skills. The prompts below are concise, professional versions of implementation requests preserved in the project evidence.

caveman — concise technical collaboration. "Report implementation progress, risks, and validation evidence in compact language without losing technical accuracy."

curate-aero-wandb-data — scientific data boundary. "Build the W&B-to-dashboard refresh and nearest-wing interaction while preserving scientific validity, deterministic provenance, physical compatibility, and protection of unpublished data."

verify-airfoil-explorer — product acceptance. "Verify that every drag preview and release keeps BP3333 geometry, C_l, C_d, C_l/C_d, and run/step provenance on one recorded row; test accessibility, responsive layout, refresh behavior, privacy, and deployment."

Observed value of the custom skill. The major improvement was not writing speed; it was eliminating plausible scientific errors in field choice, normalization, grouping, and replicate provenance before data crossed the publication boundary.

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Validation, limitations, and reflection

The deployed result is tied to current automated and live-production evidence.

Area	Evidence	Status
Exporter	59 Python tests; deterministic grouping, admission, provenance, and declared profile	**Passed**
Frontend	31 tests; ESLint; strict TypeScript; production Vite build	**Passed**
Geometry	253 finite points; Python/TypeScript max error $\approx 1.2 \times 10^{-7} < 10^{-5}$	**Passed**
Privacy	Scanner self-test and built `dist/` scan; no secret/path match	**Passed**
Custom skills	Curation and verification skill folders pass structural validation	**Passed**
Playwright	Nine browser cases passed, including axe accessibility and mobile overflow	**Passed**
Live W&B	Five reviewed runs; 2,293 admitted samples and unique geometries	**Passed**
GitHub Pages	Live workflow, artifact scan, deployment, and public HTTP checks passed	**Passed**

Physical validity and limits

Admission requires an approved finished run, a known absolute-action schema, bounded finite actions, finite positive-lift step coefficients, and a valid geometry. Runs are merged only when baseline, Reynolds number, grid, angle of attack, averaging interval, inlet condition, solver method, and immersed-boundary configuration match. Exact float32 vectors define unique geometries, while each admitted iteration remains traceable to run and step.

The deployed snapshot is live and reviewed: `snapshotKind=reviewed-wandb`, 2,293 admitted samples, 2,293 unique geometries, five source runs, and one compatible group at $Re = 3000$, grid 900×210, angle of attack 7°, and averaging window 30–60 TU. These counts are generated by the exporter rather than hard-coded into the site.

Reflection

Achieved: the public application presents the reviewed compatible design space; sliders preview the exact nearest recorded wing; C_l, C_d, and C_l/C_d update together; release snaps every parameter to that row; best-wing and NACA2412 navigation remain deterministic; and refresh validates a new snapshot atomically. AI accelerated implementation and testing, while scientific grouping and publication approval remained human decisions. **In hindsight,** I would define the public field profile, compatibility fingerprint, provenance policy, and deployment gate before the first frontend prototype.

Submission handoff

Before final submission

- Insert the second participant name, if the assignment requires one.
- Protect `main` and require CI before merge.
- Attach the final interaction video and selected screenshots.
- Recheck the public link immediately before submission.

Live Airfoil Explorer

edengol10.github.io

AI-final-course